LMRA Problem – 4 **Date : 01-04-2026**

Write the code for multiple linear regression in R for dataset : delivery and estimate the regression coeffficient, fit the model and test the hypothesis.

Solution :

```
library(robustbase)

data(delivery)

y  <- delivery$delTime
x1 <- delivery$n.prod
x2 <- delivery$distance
x  <- as.matrix(cbind(1,x1,x2))

# OLS
A <- solve(t(x)%*%x)
beta_hat <- A%*%t(x)%*%y
y_hat <- x%*%beta_hat
e <- y-y_hat

# ANOVA
mean_y <- mean(y)
y <- as.matrix(y)
SST <- t(y-mean_y)%*%(y-mean_y)
SSR <- t(y_hat-mean_y)%*%(y_hat-mean_y)
SSres <- t(y-y_hat)%*%(y-y_hat)

K <- 2
P <- 3
df2 <- length(y)-P
MSres <- SSres/df2
MSreg <- SSR/K
Fo <- MSreg/MSres

# R-square
Rsq <- SSR/SST

# t-test
t0 <- beta_hat[1]/sqrt(MSres*A[1,1])
t1 <- beta_hat[2]/sqrt(MSres*A[2,2])
t2 <- beta_hat[3]/sqrt(MSres*A[3,3])

# 95% CI beta1
tU <- qt(0.975,df2)
CI_L <- beta_hat[2]-tU*sqrt(MSres*A[2,2])
CI_U <- beta_hat[2]+tU*sqrt(MSres*A[2,2])
```

```
# ===== OUTPUT =====

beta_hat

##          [,1]
##    2.34123115
## x1 1.61590721
## x2 0.01438483

Fo

##          [,1]
## [1,] 261.2351

Rsq

##           [,1]
## [1,] 0.9595937

t0; t1; t2

##          [,1]
## [1,] 2.134738

##          [,1]
## [1,] 9.464421

##          [,1]
## [1,] 3.981313

CI_L; CI_U

##          [,1]
## [1,] 1.261825

##         [,1]
## [1,] 1.96999
```

Interpretation :

- The F-test shows the regression model is good, meaning number of products and distance together affect delivery time. R^2 tells us that most of the change in delivery time is explained by this model.
- The t-tests show both variables are important, so when number of products or distance increases, delivery time also increases. The confidence interval of β_1 not containing 0 confirms number of products has a real effect on delivery time.